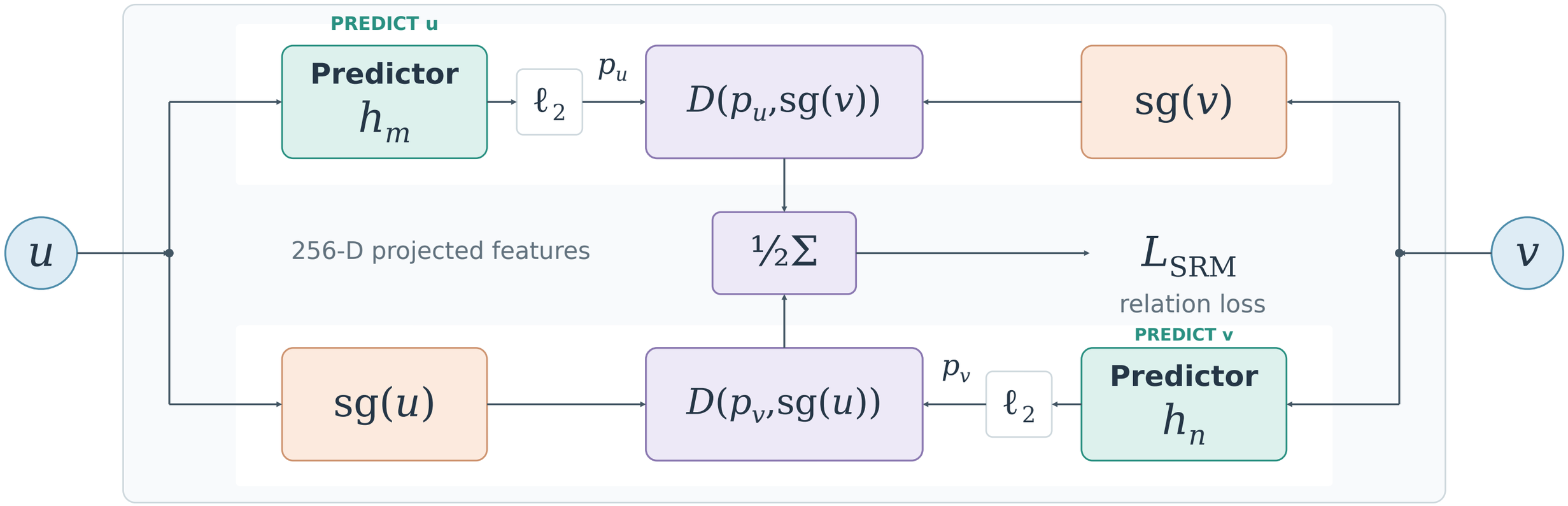
SimSiam Relation Module

SRM

Symmetric prediction • Stop-gradient targets • Shared modality predictors



$$L_{\mathrm{SRM}}=\tfrac{1}{2}[D(p_u,\mathrm{sg}(v))+D(p_v,\mathrm{sg}(u))] \qquad D(p,t)=-\cos(p,t)$$

FOUR RELATIONS · ONE REUSABLE MODULE

Positive matching only in the current mode

I–I	I–Prompt	I–Answer	Prompt–Answer
$(u,v)=(v_1,v_2)$	$(u,v)=(a^c,t_P)$	$(u,v)=(a^c,t_A)$	$(u,v)=(t_P,t_A)$
$(h_{\mathrm{I}},h_{\mathrm{I}})$ $w=1$	$(h_{\mathrm{I}},h_{\mathrm{T}})$ $w=a$	$(h_{\mathrm{I}},h_{\mathrm{T}})$ $w=c$	$(h_{\mathrm{T}},h_{\mathrm{T}})$ $w=ac$

Two predictor networks are reused by modality across all four calls. PI/PT remain outside SRM.

sg: stop-gradient ℓ_2: normalization Predictor: 256 → 128 → 256 a: human alignment c: CLIP-derived answer weight